

WLODAWA, Poland

WMO: 124970

Lat: 51.5533N

Lon: 23.5294E

Elev: 589

StdP: 14.39

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.4	6.9	-4.1	4.5	1.3	1.9	6.2	8.0	25.6	37.0	23.4	36.2	5.8	10	0.569

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	86.5	68.7	83.1	67.1	79.8	65.6	70.7	82.4	69.0	79.5	67.3	76.4	9.0	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.7	100.4	75.0	65.3	95.5	73.3	63.9	90.8	71.7	34.9	82.7	33.5	80.1	32.1	76.4	77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
21.1	18.8	16.7	-4.9	91.5	7.9	2.6	-10.6	93.4	-15.2	94.9	-19.7	96.3	-25.4	98.2	
			DB												
			WB	-5.4	73.8	7.7	1.6	-11.0	75.0	-15.5	75.9	-19.9	76.8	-25.5	78.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.9	26.5	29.2	36.3	47.6	56.7	62.5	66.7	65.3	56.4	46.9	38.2	29.9
	DBStd	16.06	10.26	9.73	8.25	7.64	6.95	6.12	5.75	5.71	6.53	7.49	8.06	9.33
	HDD50	3040	729	583	427	132	18	0	0	0	17	152	358	624
	HDD65	6834	1194	1003	889	521	271	120	49	65	268	562	803	1089
	CDD50	1926	0	0	3	61	225	375	518	475	208	55	6	0
	CDD65	245	0	0	0	1	12	45	103	75	9	0	0	0
	CDH74	2485	0	0	0	21	157	451	982	769	105	0	0	0
	CDH80	669	0	0	0	3	17	96	300	236	17	0	0	0
Wind	WSAvg	8.3	9.7	9.4	9.7	8.3	7.5	7.2	7.0	6.7	7.1	8.4	9.1	9.9
Precipitation	PrecAvg	22.3	1.2	1.1	1.3	1.6	2.5	2.6	3.3	2.5	2.0	1.5	1.4	1.3
	PrecMax	29.5	2.8	2.2	3.0	3.5	5.9	7.9	8.6	10.7	4.5	3.8	2.5	2.9
	PrecMin	15.7	0.2	0.1	0.2	0.5	0.9	0.2	0.5	0.2	0.4	0.2	0.1	0.5
	PrecStd	3.2	0.7	0.5	0.6	0.7	1.1	1.4	1.9	1.9	1.3	1.1	0.6	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.5	51.4	62.5	76.7	82.3	87.1	91.0	90.9	82.2	70.8	60.0	49.7
		MCWB	43.8	45.8	50.8	59.4	65.3	69.8	69.2	68.8	65.0	60.8	53.3	47.5
	2%	DB	43.5	47.4	56.5	70.5	78.4	83.1	87.1	86.0	76.8	66.2	55.5	45.8
		MCWB	41.1	43.6	47.2	56.3	63.5	67.2	68.7	68.6	63.1	57.6	51.4	43.9
	5%	DB	40.8	44.1	52.4	65.6	74.9	79.2	83.2	82.0	72.8	62.1	52.2	43.1
		MCWB	38.7	41.1	45.3	53.0	61.5	65.5	67.6	66.9	61.3	56.3	49.2	41.4
	10%	DB	38.3	41.0	48.3	61.7	70.8	75.9	79.6	77.9	68.5	58.3	49.0	40.7
		MCWB	36.7	38.6	43.1	51.3	59.2	63.9	66.2	64.8	59.4	53.9	46.9	39.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.1	46.8	52.0	60.3	67.7	71.4	72.9	72.5	67.0	61.9	54.6	47.7
		MCDB	46.3	50.0	61.0	75.0	78.8	83.9	85.3	85.3	78.3	68.3	58.4	49.6
	2%	WB	41.4	44.0	48.6	57.6	64.7	69.2	70.9	70.1	64.6	59.3	51.9	44.3
		MCDB	43.2	46.9	55.3	68.1	75.1	80.2	82.5	82.0	74.4	64.6	55.0	45.6
	5%	WB	39.0	41.4	45.9	54.9	62.7	67.1	69.3	68.1	62.5	56.8	49.4	41.5
		MCDB	40.5	43.9	50.8	63.3	72.7	76.2	79.6	78.5	70.4	61.2	51.8	42.9
	10%	WB	36.9	38.7	43.7	52.3	60.4	65.1	67.6	66.2	60.4	54.1	47.0	39.2
		MCDB	38.2	40.8	47.9	59.4	69.0	73.1	76.4	74.6	66.9	57.7	48.8	40.5
Mean Daily Temperature Range	5% DB	MDBR	8.4	9.9	13.1	17.8	19.2	18.9	19.2	19.9	17.5	14.4	9.1	7.9
		MCDBR	8.5	12.6	20.3	25.3	25.1	24.1	25.4	26.3	24.5	20.1	14.4	9.3
		MCWBR	7.4	9.4	12.8	13.0	12.0	11.1	9.9	10.8	12.0	12.4	10.3	7.9
	5% WB	MCDBR	8.1	12.1	18.1	22.9	23.1	21.7	21.6	22.6	21.4	18.6	12.9	9.2
	MCWBR	7.3	9.5	12.1	12.6	11.8	11.2	9.6	10.2	11.5	12.1	9.8	8.1	
Clear-Sky Solar Irradiance	taub	0.312	0.340	0.376	0.427	0.415	0.411	0.438	0.434	0.397	0.363	0.349	0.309	
	taud	2.234	2.207	2.214	2.146	2.246	2.282	2.219	2.226	2.292	2.363	2.296	2.276	
	Ebn at Noon	215	242	257	256	266	268	258	252	247	233	199	196	
	Edh at Noon	22	30	36	43	41	40	42	40	33	26	21	18	
All-Sky Solar Radiation	RadAvg	241	476	871	1304	1646	1762	1661	1478	1022	590	250	168	
	RadStd	40	68	119	155	164	142	170	113	159	96	27	30	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.95	N/A	N/A	N/A	+1.10	N/A	N/A	N/A	+133	N/A
(47)	Regional (16 neighbors)	+0.99	N/A	N/A	+1.31	+0.81	N/A	N/A	-324	+162	+67

Nomenclature: See separate page